

Yu Min Syu

Attending Physician, Department of Pediatrics,
Taipei Medical University Hospital, Taipei, Taiwan R.O.C.
Deputy Medical Officer, Center of Precision Medicine,
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Professional Summary

Board-certified Pediatrician and Clinical Geneticist with a unique blend of clinical expertise and advanced training in precision medicine and data science. A Taipei Medical University alumnus and holder of a Diploma in Precision Medicine from the University of Oxford. Currently serving as Deputy Medical Officer at the Center of Precision Medicine, specializing in AI-driven genomic analysis (GWAS/WES) and rare disease diagnostics. Dedicated to integrating machine learning applications into clinical practice to enhance diagnostic accuracy and personalized patient care.

Educational Background

Taipei Medical University, Taipei, Taiwan R.O.C.

- **Doctor of Medicine**

Master Program in Graduate Institute of Data Science

Thesis: Genetic Architecture of Body Height in the Taiwan Biobank: A Machine Learning Approach to GWAS in Underrepresented Populations

University of Oxford, St Edmund Hall, Oxford, UK

- **Diploma in Precision Medicine (07/2022)**

funded by Taiwan's National Health Research Institutes (NHRI)

Board Certification

Taiwan board of Medical Genetics and Pediatric Metabolism; Taiwan board of Adolescent and Young Adult Medicine; Taiwan board of Pediatrics

Present Position

Taipei Medical University Hospital, Taipei, Taiwan R.O.C.

Attending physician (2026 - Present)

Center of Precision Medicine, MinSheng General Hospital, Taoyuan, Taiwan R.O.C.

Deputy Medical Officer (2024 - Present)

Taipei Harvard Clinic, Taipei, Taiwan R.O.C.

Attending physician for genetic counseling (2025 - Present)

Employment Records

Mamori Pediatric Clinic, Hsinchu, Taiwan R.O.C.

Attending physician (2023 - 2025)

Division of Genetics and Metabolism, Department of Pediatrics,
Far Eastern Memorial Hospital, New Taipei City, Taiwan R.O.C.

Attending physician (2020 - 2023)

Division of Genetics and Metabolism, Department of Pediatrics,
MacKay Memorial Hospital, Taipei, Taiwan R.O.C.

Clinical Fellow in Medical Genetics (2020 - 2022)

Department of Pediatrics, National Taiwan University Hospital, Taipei, Taiwan R.O.C.

Resident (2018 - 2019)

Department of Pediatrics, Far Eastern Memorial Hospital, New Taipei City, Taiwan R.O.C.

Resident (2017 - 2020)

Publications

1. **Syu, Y. M.**, Lee, I. C., Lu, J. F., Hung, P. L., Hong, S. Y., Yang, M. T., Liang, J. S.* (2025). Insights into clinical phenotypes and treatment responses in a Small cohort of Taiwanese patients with SCN1A variants: A Preliminary study. *Pediatr Neonatol*, 66(3):223-229. <https://doi.org/10.1016/j.pedneo.2024.03.013>
2. **Syu, Y. M.***, Tung, C. Y., Chang, Y. Y., Liou, Y. J., Yang, S. C. (2025). P735: Machine learning-guided GWAS in the Taiwan biobank reveals novel height-associated genes with possible epistatic interactions. *Genetics in Medicine Open*, 3:103104. <https://doi.org/10.1016/j.gimo.2025.103104>.
3. **Syu, Y. M.**, Lee, C. L., Chuang, C. K., Chiu, H. C., Chang, Y. H., Lin, H. Y.*, & Lin, S. P.* (2022). Functional Independence of Taiwanese Children with Osteogenesis Imperfecta. *Journal of personalized medicine*, 12(8), 1205. <https://doi.org/10.3390/jpm12081205>
4. **Syu, Y. M.**, Lee, H. C., Chang, J. H., Lee, C. L., Chuang, C. K., Chiu, H. C., Chang, Y. H., Lin, H. Y.*, & Lin, S. P.* (2022). Rapid Weight Loss and Severe Failure to Thrive Mimicking Lipodystrophy Syndrome in a 1-Year-Old Taiwanese Girl with Costello Syndrome. *Children (Basel, Switzerland)*, 9(6), 905. <https://doi.org/10.3390/children9060905>
5. **Syu, Y. M.**, Ma, J. Y., Ou, T. H., Lee, C. L., Lin, H. Y., Lin, S. P., Lee, C. J.*, Chen, C. P.* De Novo Mosaic 6p23-p25.3 Tetrasomy Caused by a Small Supernumerary Marker Chromosome Presenting Trisomy Distal 6p Phenotype: A Case Report and Literature Review. *Diagnostics* 2022, 12, 2306. <https://doi.org/10.3390/diagnostics12102306>
6. Lee, C. L., Chuang, C. K., **Syu, Y. M.**, Chiu, H. C., Tu, Y. R., Lo, Y. T., Chang, Y. H., Lin, H. Y.*, Lin, S. P.* Efficacy of Intravenous Elosulfase Alfa for Mucopolysaccharidosis Type IVA: A Systematic Review and Meta-Analysis. *J. Pers. Med.* 2022, 12, 1338. <https://doi.org/10.3390/jpm12081338>

7. Lee, C. L., Lin, S. M., Chen, M. R., Chuang, C. K., **Syu, Y. M.**, Chiu, H. C., Tu, R. Y., Lo, Y. T., Chang, Y. H., Lin, H. Y.*, & Lin, S. P.* (2022). Long-Term Cardiovascular Findings in Williams Syndrome: A Single Medical Center Experience in Taiwan. *Journal of personalized medicine*, 12(5), 817. <https://doi.org/10.3390/jpm12050817>

Conference Presentations

1. **Yu Min Syu**, Min Huei Hsu, Yi-Hsin (Elsa) Hsu. (2024). Mapping the Impact of Next-Generation Sequencing: A 20-Year Bibliometric Study on Key Shifts and Emerging Frontiers in Precision Medicine. Paper presented at The 2024 Multiomics and Precision Medicine Joint Conference, Taipei, Taiwan R.O.C.
2. **Yu Min Syu**, Chung-Lin Lee, Hsiang-Yu Lin, Shuan-Pei Lin. (2021). A 10-year-old girl presented with hypotonia in early childhood and progressive muscle atrophy. Oral presentation at The 2021 Autumn Symposium of the Taiwan Human Genetics Society, Taichung, Taiwan R.O.C.
3. **Yu Min Syu**, Jao Shwann Liang. (2019). Perampanel as effective add-on antiepileptic drug for refractory epilepsy in a patient with *MECP2* duplication syndrome. Paper presented at The 20th Annual Meeting of Infantile Seizure Society, Nagoya, Japan.
4. **Yu Min Syu**, Jyh Feng Lu, Jao Shwann Liang. (2018). Broaden the Phenotype of *SCN1A* Channelopathies: Results of Taiwanese cohort study. Paper presented at the International Symposium on Severe Infantile Epilepsy Treatment, Rome, Italy.

Writing and Media Engagement

- Columnist for Taiwan's largest parenting magazine, Parenting (親子天下) (2023 - Present)
- 2025 Columnist of the year, Parenting (親子天下)
- Author of health education book: "Growing Taller, Not Heavier: A Pediatric Expert's Guide to Building a Foundation for Height Acceleration and Healthy Body Development" (Global Publishing Group, Feb 2025)

Research Interest

Phenotype extraction system from EMR, Genome-wide association study, machine learning application in genomic study, genomic data and healthcare record governing, variable expressivity of disease-causing genes, phenotypical variability in genetic epilepsy, neurodevelopmental and rare genetic diseases

Research Activities

Center of Precision Medicine, MinSheng General Hospital, Taoyuan, Taiwan R.O.C.

Deputy Medical Officer and Principal Investigator (2024 - present)

- Developing machine learning-enhanced genome-wide association study using Taiwan Biobank Dataset and MinSheng Dataset
- Established analytic algorithm of whole exome sequencing using NGS technique

- Leading Laboratory-Developed Tests (LDTs) genetic testing certification team

Department of Pediatrics, Far Eastern Memorial Hospital, New Taipei City, Taiwan R.O.C.

Principal Investigator (2021 - 2023)

- Conducting research project: Genotype, phenotype, natural history and functional independence of Taiwanese children with genetic epilepsy
- Established analytic algorithm of whole exome sequencing and mitochondria DNA sequencing using NGS technique

Research Fellow (2018 - 2020)

Research advisor: Dr. Jao-Shwann Liang

- Conducted research project: Clinical Manifestations and Outcomes of Patients with *SCN1A* Channelopathy
- Managed analysis of whole exome sequencing results and interpretation of pathogenicity

Division of Genetics and Metabolism, Department of Pediatrics, MacKay Memorial Hospital, Taipei, Taiwan R.O.C.

Research Fellow (2020 - Present)

Research advisor: Prof. Shuan-Pei Lin. Second advisor: Hsiang-Yu Lin.

- Conducting research project: “The evaluation of genotype, phenotype, natural history and cardiac function in patients with genetic rare diseases.” Focusing mainly on cardiac diastolic functional changes in patients with connective tissue diseases, e.g., osteogenesis imperfecta.
- Conducting research project: “Functional independence of Taiwanese children with rare diseases.” Focusing mainly on prognosis of children skeletal and neurodevelopmental diseases, e.g., achondrodysplasia and Prader–Willi syndrome.
- Conducted and managed research project: Clinical Manifestations and Genotypes of Patients with Costello syndrome.”
- Conducted research in mosaic partial 6p tetrasomy and its clinical presentations
- Providing genetic counseling interpretation of pathogenicity for patients with whole exome sequencing results

Core Laboratory of Biomedical Mass Spectrometry Imaging, Office of R&D Taipei, Taiwan R.O.C.

Graduate Institute of Medical Sciences, Taipei Medical University Taipei, Taiwan R.O.C.

Undergraduate Research Fellow (2010 – 2012)

Research advisor: Dr. Jung-Shan, Chang

- Identified psoriasin overexpression in human breast cancer cell as a biomarker
- Assisted establishment of a new model demonstrating leukocyte sequestration in pulmonary microvessels in a murine ovalbumin model of asthma using Intravital Microscopy